

CREDIT RISK EVALUATION USING SUPPORT VECTOR MACHINE

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Abstract

Credit risk or also known as bad credit risk is defined as the risk that occurs due to the inability of the customer to repay the loan and the interest within a certain period of time. Bad credit can be prevented by selecting proper customers during the loan application process so as not to give losses to the bank.

This study aims to analyze credit risk using machine learning methods. One of the machine learning algorithms that can be used for data classification is Support Vector Machine(SVM). The concept of SVM is to find optimal hyperplane with maximum margin to linearly separate the data set into two classes.

The dataset were taken from kaggle named as *german_credit_data.csv*. Dataset was divided into two parts along with percentage of training data 80% and the testing data 20%. The variables used were age, sex, job, housing, saving accounts, checking account, credit amount, duration and purpose.

Thus, this model can be used to classify prospective customers into good credit or bad credit class with sufficiently high accuracy so as to help banks reduce the risk of bad credit.

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1 Introduction

The support vector machine, an approach for classification that was developed in the computer science community in the 1990s and that has grown in popularity since then. Most research with machine learning forecasting has focused on Artificial Neural Networks (ANN). Recent research in the field has used another technique known as Support Vector Machines (SVMs) in addition to or as an alternative to ANNs. Whereas ANNs are models that try to minimize classification error within the training data, SVMs may make classification errors within training data in order to minimize overall error across test data. A major advantage of SVMs is that it finds a global optimum, whereas neural networks may only find a local optimum. A support vector machine can be imagined as a surface that creates a boundary between points of data plotted in a multidimensional space representing examples and their feature values. The goal of an SVM is to create a flat boundary called a hyperplane, which divides the space to create fairly homogeneous partitions on either side. The concept of the support vector classifier (SVC) is similar to another popular linear regression model, Ordinary Least Squares (OLS), but the two optimize different quantities. The OLS finds the residuals, the distance from each data point to the fit line, and minimizes the sum of squares of these residuals. The SVC on the other hand looks at only the support vectors, and uses the inner product to maximize the distance from hyperplane to the support vector. Additionally, the inner products in SVC are weighted by their labels, whereas in OLS the square of residuals serves as the weighting. Thus SVC and OLS are

two different methods of approaching this problem. The machine conceptually implements the following idea: input vectors are non-linearly mapped to a very high dimension feature space. In this feature space a linear decision surface is constructed. Special properties of the decision surface ensures high generalization ability of the machine learning.

2 Machine Learning

Machine learning is a branch of artificial intelligence and computer science that focuses on analyzing and interpreting patterns and structures in data to enable learning, reasoning, and decision making outside of human interaction. It is being used for various tasks such as image recognition, internet search engines, email filters to sort out spam, speech recognition, Facebook auto-tagging, recommender system, and many more.

In the real world we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from experiences or past data like a human does? So here comes the role of **Machine Learning**. Machine Learning is about designing algorithm that automatically extract valuable information from data. The emphasis here is on “automatic”, i.e., machine learning is concerned about general-purpose methodologies that can be applied to many datasets, while producing something that is meaningful.

Types of Machine Learning

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Reinforcement Machine Learning

2.1 Supervised Learning

Algorithms that learn from training datasets and are being trained to monitor learning are referred to by the phrase “Supervised Learning”. The algorithm picks up the mapping function so it may generate an output from the input. To forecast the results for fresh input data, the learned mapping function is then used. The algorithm reaches the desired degree of performance for result prediction after several iterations. All input data must be labeled because classes used in supervised learning are known. Supervised learning aims to locate data clusters. When we want to do classification (map input labels to output labels) or regression (map input labels to continuous output labels), we often employ supervised learning. A model is trained using supervised learning to perform regression or classification. As a result, supervised learning algorithms are divided into groups for classification and regression:

Based on training data, the **CLASSIFICATION** algorithm is a supervised learning method that is used to categorize new observations. Classification produces categories, not values, such as “Green or Blue”, “spam or not spam” as its output variable.

When the output variable has a real or continuous value, like a salary or weight, the situation is known as a **REGRESSION** problem.

In both situations, the objective is to identify a certain relationship or structure among the input variables that will aid in producing and correcting output. Based on the training data, the output is chosen. Because the input variables are known and labeled, which implies that the algorithm will only try to find the hidden patterns, supervised learning generates accurate and trustworthy results.

2.2 Unsupervised Learning

The input data for unsupervised machine learning is unlabeled or uncategorized. In contrast to supervised learning, the number of classes is unknown, and no prior knowledge of the data exists. Unsupervised learning algorithms simulate the distribution of the input data without knowing the labels. Before the algorithm can find hidden patterns and structures, the input data must be defined and labeled. The data should be clustered using a variety of categorize, as predicted. In clustering and association groups, these algorithms are categorised as follows:

The task of clustering is grouping the population or data into a variety of groups so that data points in the same groups are more comparable to other data points in the same group and dissimilar to the data points in other groups. It is essentially a grouping of objects based on how similar and unlike they are to one another.

Unsupervised learning techniques such as association rule learning look for dependencies between data items and map the data accordingly to maximize profitability.

Clustering is the most common unsupervised technique used to find hidden patterns or create clusters. Unsupervised learning is used

to explore the data and train a model to split data up into clusters.

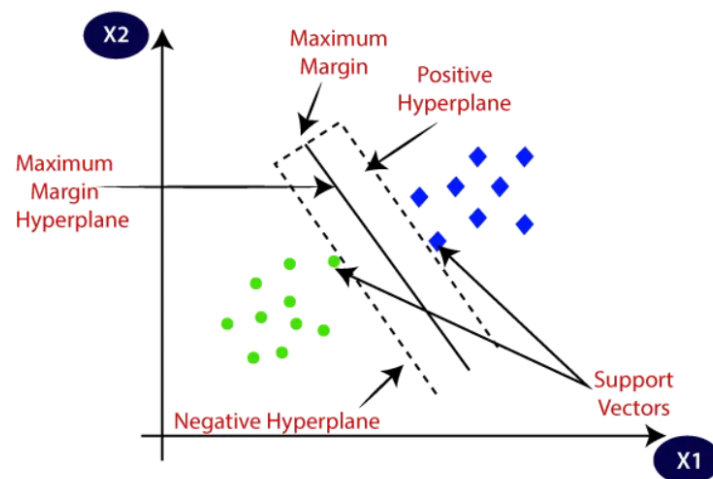
2.3 Reinforcement Learning

Reinforcement machine learning algorithm is a learning method that interacts with the environment by producing actions and discovering errors. Trial, error, and delay are the most relevant characteristics of reinforcement learning. In this technique, the model keeps on increasing its performance using Reward Feedback to learn the behavior or pattern. These algorithms are specific to a particular problem e.g. Google Self Driving car, AlphaGo where a bot competes with humans and even itself to get better and better performers in Go Game. Each time we feed in data, they learn and add the data to their knowledge which is training data. So, the more it learns the better it gets trained and hence experienced.

3 Preliminaries

- **Hyperplane:** A decision boundary separating different classes in feature space and is represented by the equation $w^T x + b = 0$ in linear classification.
- **Support Vectors:** The closest data points to the hyperplane or the points that are actually passing through the hyperplane.
- **Margin:** The distance between the hyperplane and the support vectors. SVM aims to maximize this margin for better classification performance.

- **Kernel:** A function that maps data to a higher-dimensional space enabling SVM to handle non-linearly separable data.
- **Hard Margin:** A maximum-margin hyperplane that perfectly separates the data without misclassifications.
- **Soft Margin:** Allows some misclassifications by introducing slack variables, balancing margin maximization and misclassification penalties when data is not perfectly separable.



4 Support Vector Machine

A support vector machine is a supervised machine learning algorithm that can be used for classification and regression tasks. The main idea behind SVM is to find the best boundary (or hyperplane) that separates the data into different classes.

In the case of classification, an SVM algorithm finds the best boundary that separates the data into different classes and the boundary is chosen in such a way that it maximizes the margin.

SVM can also be used for non-linear classification by using a technique called the kernel trick. The kernel trick maps the input data into a higher-dimensional space where the data becomes linearly separable.

5 Maximum Margin Classifier

Hyperplane: A hyperplane H is a linear subspace of a vector space V such that the basis of H has cardinality one less than the cardinality of the basis for V . In other words, if V is an n -dimensional vector space then H is an $(n - 1)$ -dimensional flat affine space.

A hyperplane always divides the parent vector space into two regions. For example, in two dimensions, a hyperplane is a line. In three dimensions, a hyperplane is a flat two-dimensional subspace—that is, a plane. In $p > 3$ dimensions, it can be hard to visualize a hyperplane, but the notion of a $(p - 1)$ -dimensional flat subspace still applies. In p -dimensional setting, equation of p -dimensional hyperplane is:

$$w_0 + w_1X_1 + w_2X_2 + \dots + w_pX_p = 0$$

where $X = (X_1, X_2, \dots, X_p) \in R^p$ satisfies the hyperplane equation implies lies on the hyperplane.

Now, suppose that X does not satisfy above equation, then

$$w_0 + w_1X_1 + w_2X_2 + \dots + w_pX_p > 0$$

This tells us that X lies to one side of the hyperplane. On the other hand, if

$$w_0 + w_1X_1 + w_2X_2 + \dots + w_pX_p < 0$$

then X lies on the other side of the hyperplane.

6 Mathematical Computation

Consider a binary classification problem with two classes, labeled as +1 and -1. We have a training dataset consisting of input feature vector X and their corresponding class labels Y . The equation for the linear hyperplane can be written as:

$$w^T x + b = 0$$

where w is the normal vector to the hyperplane and b is the offset or the term that represents the distance of the hyperplane and the origin along the normal vector w .

Distance from a data point to the hyperplane

The distance between a data point x_i and the decision boundary can be calculated as

$$d_i = \frac{w^T x_i + b}{||w||}$$

where $||w||$ is the Euclidean norm of the weight vector w .

Linear SVM Classifier

Distance from a Data point to the Hyperplane:

$$y_i = \begin{cases} 1, & \text{if } w^T x + b \geq 0 \\ -1, & \text{if } w^T x + b < 0 \end{cases} \quad (1)$$

It can be also written as:

$$y_i(w^T x_i + b_i) \geq 1$$

where y_i is the predicted label of a data point.

Optimization Problem for Linear SVM is:

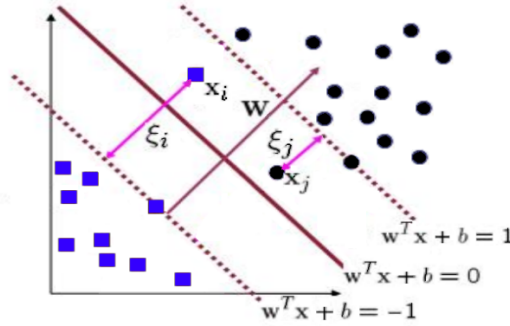
$$\text{Minimize } \frac{1}{2} ||w||^2$$

subject to the constraint:

$$y_i(w^T x_i + b_i) \geq 1 \text{ for } i = 1, 2, \dots, m$$

where, y_i is the class label for each training instance, x_i is the feature vector of the i^{th} training instance and m is the total number of training instances.

If the training samples are not linearly separable, then the maximum-margin hyperplane does not exist. Then the SVM formulation can be extended to non-separable cases based on a concept called the **soft margin**.



The soft SVM formulation aims to optimize a linear combination of the two; namely, it tries to maximize the margin as much as possible and simultaneously tries to minimize the total introduced errors as well.

In the soft margin for SVM, the data points which are on the incorrect side of the margin have a penalty term that increases with the distance from it.

So our objective function is

$$\min \frac{1}{2} \|w\|^2 + R \sum_{i=1}^m \zeta_i$$

subject to constraints

$$y_i(w^T x_i + b_i) - 1 + \zeta_i \geq 0 \quad \forall i$$

The trade-off between the slack variable penalty and the amount of the margin is controlled by parameter R .

7 Non Linear Support Vector Machine

The key idea of nonlinear SVMs is that we first select a nonlinear function $h(x)$ to map each input x_i into $h(x_i)$ in a higher-dimensional space, and then we follow the same SVM procedure. Then we use the kernel trick for mapping the nonlinear function into higher-dimensional space.

The kernel function is a function mapping from two inputs in the low-dimension space to a real number in R ; thus, it is much more convenient to specify the kernel function than the mapping function itself. We can specify any function as the kernel function $\phi(x_i, x_j)$ as long as it satisfies the Mercer's condition.

1. Linear kernel:

$$\phi(x_i, x_j) = x_i^T x_j,$$

which corresponds to an identity-mapping function.

2. Polynomial kernel:

$$\phi(x_i, x_j) = (x_i^T x_j)^p,$$

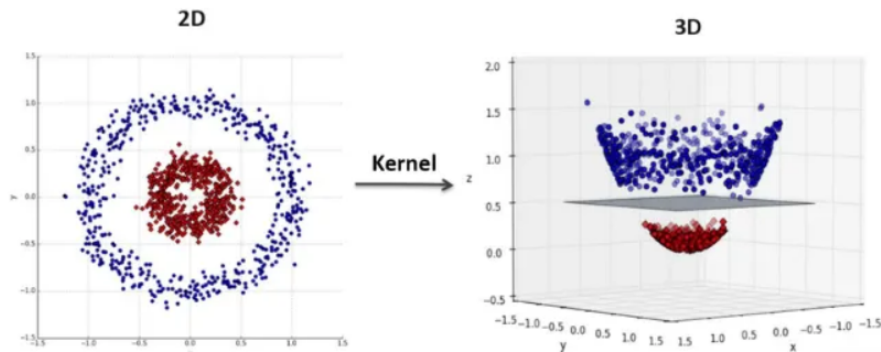
where p is the order of the polynomial.

3. Gaussian(or RBF) kernel:

$$\phi(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2),$$

where γ is a hyperparameter to control the variance of the Gaussian.

Here, in this there is non linear data and we have to fit the hyperplane in this data so we use the polynomial kernel to fit the hyperplane in the data and classifies the data into classes.



Choosing the Right Kernel

Selecting the appropriate kernel for your SVM model depends on several factors:

- **Data Complexity:** For linearly separable data, the linear kernel is sufficient. For more complex data, consider polynomial or RBF kernels.

- **Computational Resources:** RBF and polynomial kernels are computationally more intensive than the linear kernel. Ensure that your computational resources can handle the increased complexity.
- **Model Performance:** Experiment with different kernels and use cross-validation to determine which kernel yields the best performance for your specific problem.

7.1 Advantages of Support Vector Machine (SVM)

- **High-Dimensional Performance:** SVM excels in high-dimensional spaces, making it suitable for image classification and gene expression analysis. **Nonlinear Capability:** Utilizing kernel functions like RBF and polynomial SVM effectively handles nonlinear relationships.
- **Outlier Resilience:** The soft margin feature allows SVM to ignore outliers, enhancing robustness in spam detection and anomaly detection. **Binary and Multiclass Support:** SVM is effective for both binary classification and multiclass classification suitable for applications in text classification.
- **Memory Efficiency:** It focuses on support vectors making it memory efficient compared to other algorithms.

7.2 Disadvantages of Support Vector Machine (SVM)

- **Slow Training:** SVM can be slow for large datasets, affecting performance in SVM in data mining tasks. **Parameter Tuning**

Difficulty: Selecting the right kernel and adjusting parameters like C requires careful tuning, impacting SVM algorithms.

- **Noise Sensitivity:** SVM struggles with noisy datasets and overlapping classes, limiting effectiveness in real-world scenarios. Limited Interpretability: The complexity of the hyperplane in higher dimensions makes SVM less interpretable than other models.
- **Feature Scaling Sensitivity:** Proper feature scaling is essential, otherwise SVM models may perform poorly.

8 Credit Risk Evaluation Model

Credit risk refers to the possibility that a borrower will fail to meet their loan obligations. In simple terms -it's the risk of default.

The goal of credit risk evaluation is to build a predictive model that classifies applicants into:

- Low risk (good credit) → likely to repay the loan
- High risk (bad credit) → likely to default

Machine learning models help financial institutions automate this process, improve accuracy, and reduce human bias.

- The classification process of dataset was done by using Python. The dataset was taken from the kaggle named as *german_credit_data.csv* and with up to 1000 data and 9 variables.
- The dataset were divided into two parts along with percentage of the training data 80% and the testing data 20%. Model was made using the training data then checked using the testing data to discover its performance.
- Model evaluation was counted using confusion matrix which are classification report, accuracy, accuracy score.

8.1 Python Code

Import all the required libraries and read all the data with the help of the command *pd.read_csv*.

```
1 import pandas as pd
2 import numpy as np
3 from sklearn.model_selection import train_test_split, GridSearchCV
4 from sklearn.preprocessing import StandardScaler, LabelEncoder
5 from sklearn.svm import SVC
6 from sklearn.metrics import classification_report, confusion_matrix,
    accuracy_score
7 data = pd.read_csv("german_credit_data.csv")
```

Encode categorical variables

```
1 for col in data.select_dtypes(include=['object']).columns:
2     le = LabelEncoder()
3     data[col] = le.fit_transform(data[col])
4     #Create a new column for risk
5 data['risk']=((data['Credit amount']>5000) & (data['Duration']>24)).
    astype(int)
6 # Features & target
7 X = data.drop("risk", axis=1)    # Risk = target column
8 y = data["risk"]
9 # Train-test split
10 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size
    =0.2, random_state=42)
11 # Standardization
12 scaler = StandardScaler()
13 X_train = scaler.fit_transform(X_train)
14 X_test = scaler.transform(X_test)
```

Build and train the SVM model

```
1 model_svm=SVC()
2 model_svm.fit(X_train,y_train)
3 #Make prediction and evaluate the SVM model
4 svm_predictions=model_svm.predict(X_test)
5 accuracy_score_svm=accuracy_score(svm_predictions,y_test)
6 confusion_matrix_svm=confusion_matrix(svm_predictions,y_test)
```

```

7 classification_report_svm=classification_report(svm_predictions,
  y_test)
8 print("SVM Model:")
9 print("Confusion Matrix:\n", confusion_matrix_svm)
10 print("\nClassification Report:\n", classification_report_svm)
11 print("\nAccuracy Score:", accuracy_score_svm)

```

This is the output of the SVM model:

```

SVM Model:
Confusion Matrix:
[[176  5]
 [ 0 19]]

Classification Report:
              precision    recall  f1-score   support

     0           1.00       0.97       0.99         181
     1           0.79       1.00       0.88          19

   accuracy                0.97         200
  macro avg           0.90       0.99       0.93         200
 weighted avg           0.98       0.97       0.98         200

Accuracy Score: 0.975

```

Table 1: Performance Metrics for Credit Risk Evaluation Model

Metric	Formula	Meaning
Accuracy	$(TP + TN) / \text{Total}$	Overall correctness
Precision	$TP / (TP + FP)$	How many predicted defaulters actually defaulted
Recall (Sensitivity)	$TP / (TP + FN)$	How many actual defaulters were detected
F1-Score	$2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$	Balance between precision and recall
Confusion Matrix	Table showing classifications	Visual representation of model performance

9 Conclusion

In the output of the credit risk evaluation model we can see that it predicts the accurate result for 195 data from 200 data. Therefore, this model gives a 97.5% times correct result and the accuracy of the model is 0.975. This means that the SVM model can classify the data well.

This model can be used to assist bank in choosing the decision to accept or reject a credit application by classifying the prospective customer data into good credit or bad credit class. So that bank can reduce the risk of loss due to bad credit.

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